

Computing Methods for Physics – 6 February 2023

Your exam material (code files, plots, etc.) must be submitted via google classroom by 13:00 as a single zip file.

C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, setters/getters, unnecessary void functions, correct mathematical expression and physical units, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling and clarity in general.

Part 1 – C++

1. A capacitor is characterized by its capacitance $C = Q/\Delta V$ — measured in Farad [F] — which regulates the ratio between the charge $Q > 0$ on its positively charged plate and the differential potential ΔV between its two plates. These quantities can be used to determine the energy stored in the capacitor:

$$U = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}Q\Delta V = \frac{1}{2}\frac{Q^2}{C}. \quad (1)$$

Implement a class `Capacitor` to represent spherical capacitors, for which

$$C = 4\pi\epsilon_0\epsilon_r \frac{R_1 R_2}{R_2 - R_1}, \quad (2)$$

where $\epsilon_0 = 8.854 \cdot 10^{-12}$ F/m is the vacuum permittivity, R_1 and R_2 are the radii of the inner and the outer spherical plate, respectively, and $\epsilon_r \geq 1$ is the relative permittivity of the material between the plates ($\epsilon_r = 1$ for vacuum). The class must provide 3 constructors so that the user can create instances of these capacitors by providing Q , ΔV , or U in addition to R_1 , R_2 , and ϵ_r .

2. Implement a class `CompositeCapacitor` which inherits from `Capacitor` and allows the user to represent spherical capacitors in which the gap between the plates is filled with N layers of different materials (so different ϵ_r 's) with the same thickness. The total capacitance of a this object is given by

$$C = (C_1^{-1} + \dots + C_N^{-1})^{-1}, \quad (3)$$

where C_i is the capacitance of layer i . Also this class must provide the user with 3 constructors, but note that this time ϵ_r will not be a single value: instead $\vec{\epsilon}_r = \{\epsilon_{r,1}, \dots, \epsilon_{r,N}\}$.

3. Write an application `app.cpp` to showcase the methods of your classes.

Part 2 – Python

Use a Python notebook or Python scripts to complete the following tasks.

1. Load the dictionary stored in the file `data.pickle`.
2. Plot the data content of the dictionary.
3. Set up a Monte Carlo with `emcee` to fit the data.
4. Display and comment the results of your fit.